

COMP47670 Sample Exam - Autumn 2025/26

Guidelines:

- Complete all three tasks. All tasks carry equal marks.
- All tasks should be completed in a single Jupyter Notebook (IPYNB file). A template Jupyter notebook file has been provided on Brightspace.
- Use Markdown cells to briefly explain your work.
- Limit of 2 hours allowed for the exam. Once time has elapsed, you must submit your Jupyter notebook. Please monitor the clock to ensure you are not late.
- No collaboration is allowed. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- This is an open book exam (i.e., you can use module lecture/lab material and internet resources).
- Generative AI tools, code assistants, and large language models (such as ChatGPT) must not be used during the exam. Development environments such as VSCode or Colab also must not be used for the exam.

Summary:

This exam involves using Python to analyse a dataset containing records of car sales from dealerships in different parts of Ireland during a one-year period. Each row in the data represents the sale of a single car, described by the following features:

- *date*: the date when the car sale occurred
- *make*: the make or manufacturer of the car
- *model*: the specific model name of the car
- *year*: indicates the age of the car
- *transmission*: indicates if the car is manual, automatic, or semi-automatic
- *fuel_type*: specifies the type type of fuel used by the car
- *mileage*: the distance (in kilometres) that the car has previously been driven
- *region*: the province in Ireland where the car sale took place
- *sale_amount*: the amount (in euros) for which the car was sold

Task 1: Data Preparation

- a) Download and load the car sales dataset, which is stored in tab-separated format at the following link:
<http://mlg.ucd.ie/modules/python/sample/cars.tsv>
- b) Report the percentage of rows with missing values and remove these rows from the dataset.
- c) Examine the extent to which there are features with irregular cardinality in the data. Apply appropriate preprocessing steps to address this issue.

Task 2: Feature Exploration

- a) Produce a plot showing how the total sale amount for cars sold in the full dataset relates to the region in which the sale took place.
- b) Investigate the relationship between the model and the transmission of the cars in the full dataset.
- c) Create a plot showing illustrating the age of the cars in the dataset.

Task 3: Time Series Analysis

- a) Filter the dataset to only include cars sold in either the Leinster or Munster region. Report the percentage of rows retained in the filtered dataset.
- b) Using the filtered data from Task 3a, create time series showing the monthly number of sales for each fuel type. Visualise all three time series on a single plot.
- c) Briefly comment on the trends appearing in the plots from Task 3b.